

Agentic AI in Healthcare: Opportunities, Challenges, and Future Directions

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The integration of Agentic AI into healthcare marks a new era in medical automation and decision support. Leveraging autonomous reasoning, planning, and collaboration, these AI systems are transforming clinical workflows, diagnostics, and patient management. Recent advancements have led Agentic AI to evolve from single-agent decision-making to multi-agent systems, enabling more sophisticated problem-solving in complex medical environments. In this survey, we provide an in-depth discussion on the core aspects and challenges of Agentic AI in healthcare, including its applications in electronic health records (EHR) interactions, clinical triage, medical question-answering, and disease diagnosis. We categorize existing single-agent and multi-agent frameworks, explore key evaluation metrics, implementation strategies, and commonly used benchmarks, and examine the communication, reasoning, and decision-making processes of these AI agents. Additionally, we address critical challenges such as model reliability, interoperability, ethical considerations, and regulatory constraints. To support further research, we summarize relevant datasets and benchmarks and we outline future research directions that emphasize human-AI collaboration, transparency, and the safe deployment of AI-driven medical systems.

CCS Concepts: • **Agentic AI**; • **Large Language Models**; • **Healthcare**;

Additional Key Words and Phrases: AI Agents, LLMs, Natural Language Processing, Survey

1 Introduction

Agentic AI, also called compound AI systems, has recently gained significant attention in the research community due to its potential to enhance autonomy, adaptability, and collaboration in various applications. In a recent article, [138] argued that “Compound AI systems will likely be the best way to maximize AI results in the future and might be one of the most impactful trends in AI in 2025”. To solve a problem, AI agents rely on four steps: perception, reasoning, acting, and learning. In Agentic AI, we can distinguish between single-based agents and multi-based agents’ systems, where the latter has gained much attention because their ability to solve many complex problems and provide advanced capabilities by allowing each agent to focus on a single task

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ACM 1557-7341/2026/4-ART

<https://doi.org/10.1145/3809164>

and achieve it effectively and accurately. Additionally, by relying on multiple agents and interactions between them, one can simulate complex real-world environments efficiently. Therefore, many open-source frameworks have been released, covering areas such as tool using [99], web interaction [144], programming [44, 48, 89], multi-robot systems [70, 140], society simulation [84, 85], among others. Furthermore, Numerous benchmarks initially designed for LLM evaluation have also been adapted for assessing AI agents.

Thanks to its interdisciplinary nature, the field of Agentic AI has drawn interest from a wide range of researchers, extending beyond AI specialists to include experts from healthcare, social sciences, games, psychology, policy studies, among others. Recent innovations in Agentic AI include frameworks that enhance autonomy and continuous learning, enabling agents to execute tasks independently while adapting to dynamic environments. These advancements are collectively pushing the boundaries of Agentic autonomy and adaptability to facilitate multi-agent communication, task orchestration, and integration with external APIs, empowering agents to perform complex, multi-step tasks autonomously.

While recent progress in Agentic AI has been largely driven by advances in LLMs and open-source tooling, the idea of autonomous agents in healthcare has deep historical roots. Foundational systems such as MYCIN [106], INTERNIST-I [73], and CADUCEUS [88] pioneered rule-based expert systems designed to support clinical decision-making in areas like infectious diseases and internal medicine. These early systems used symbolic reasoning, handcrafted rules, and inference engines to mimic human diagnostic processes—often achieving performance comparable to that of human specialists. Though limited by the computational and data constraints of their time, they laid the intellectual groundwork for many of the agentic architectures that followed, particularly in their emphasis on explainability, modularity, and domain-specific expertise.

In healthcare, Agentic AI represents a significant progress, offering autonomous, adaptive, and collaborative systems capable of executing complex tasks with minimal human intervention. Therefore, agents have the potential to revolutionize the healthcare domain when it comes to patient care, streamline operations, and accelerating research. Agentic AI is unlocking a wide range of applications in healthcare, addressing diagnostics [17, 31, 54, 58, 96, 146], handling EHR [105, 124], medical question answering, [54, 59, 101] telemedicine, and remote monitoring [5], mental health [32, 128, 142] and reasoning [43, 54, 115, 123]. With the explosion of open-source frameworks such as LangChain¹, CrewAI², notably, along with publicly available datasets and the advancement in multi-modal learning, Agentic AI systems are well positioned to solve complex challenges in healthcare.

While Agentic AI holds great promise in healthcare, it also amplifies known AI risks and introduces new ones, given the speed, scale, and autonomy these systems enable. These risks reflect the profound impact that healthcare decisions can have on people's lives. Key concerns include errors in decision-making, lack of accountability, trust and privacy issues, bias and discrimination, misuse of data or intellectual property, and insufficient human oversight. As AI applications in healthcare are still in their infancy, addressing these challenges is crucial to ensuring the safe, responsible, and effective deployment of Agentic AI across all levels of healthcare. Several recent works have begun to explore the intersection of Agentic AI and healthcare. Early surveys on conversational agents and agent-based systems in healthcare provided foundational insights into their use cases, design, and implementation challenges [75, 110]. More recent efforts have proposed ethical and clinical frameworks for agentic reasoning in medical contexts [97]. Additionally, within the multi-agent literature, growing attention is being paid to the failure modes, coordination bottlenecks, and risks associated with scaling multi-agent LLM systems [11, 37, 38], all of which carry critical implications for safety and reliability in healthcare deployments. In the past two years, numerous survey papers have been published on Agentic AI [21, 38, 60, 71, 127], however we have identified several limitations in these works: (1) the major literature review focused on general Agentic AI

¹<https://www.langchain.com/>

²<https://www.crewai.com/>

with applications to various domains [18, 21, 33, 120, 127]; or (2) other specific domains such as gaming [45, 130], social intelligence [72], finance [20], and robotics [130]; or (3) largely focused on LM-based agents [14, 34, 131]. Consequently, there is a lack of research on the applications and risks of autonomous agents in healthcare, leaving an absence of systematic reviews addressing their impact, challenges, and future directions. Our work fills this gap by providing a dedicated survey on Agentic AI in healthcare, examining its potential to transform patient care, streamline administrative processes, and enhance overall healthcare outcomes.

A concurrent survey paper has recently been proposed exploring LLM-based agents in medicine [119]. While both surveys explore the role of LLM-based agents in medicine, they differ significantly in scope, focus, and analytical depth. [119] provides a broad overview of LLM-driven medical agents, categorizing them based on their architectures, external tool integrations, and clinical applications. It examines how these agents assist in clinical decision support, medical documentation, training simulations, and healthcare service optimization. In contrast, our survey takes a more specialized approach, focusing on Agentic AI in healthcare, which goes beyond standard LLM automation to explore autonomous, adaptive, and collaborative AI systems. It systematically differentiates single-agent and multi-agent frameworks, analyzing their reasoning capabilities, autonomy in medical workflows, and ethical considerations. Furthermore, our survey delves deeper into interpretability, security, privacy, and real-world scalability, addressing critical challenges that must be overcome for the safe and trustworthy deployment of Agentic AI in medicine. By shifting the perspective from task-specific LLM applications to fully autonomous medical AI systems, this work provides a forward-looking analysis on the future of Agentic AI in healthcare.

In preparing this survey, we followed a focused literature selection process aimed at capturing the most relevant and recent developments in Agentic AI applied to healthcare. Our review primarily covers works published between 2020 and mid-2025, reflecting the rapid growth of the field during this period. We included peer-reviewed papers, preprints, and key open-source frameworks that introduced agentic behaviors such as autonomy, memory persistence, multi-agent collaboration, and tool integration. Foundational systems were also included to provide historical context. Papers were selected based on their relevance, novelty, and contribution to the taxonomy and themes developed throughout the survey. The selection of papers was finalized in February 2025, which corresponds to the date of submission of this manuscript for review.

2 Agentic AI: Foundations and Key Concepts

2.1 Definition and Characteristics

The concept of “agent” has a rich history and has been explored across various disciplines including philosophy. It has been discussed by influential philosophers starting from Aristotle to Hume among others. Generally, an “agent” is an entity that has the ability to act, while the concept of “agency” refers to the exercise or representation of this ability [100]. In artificial intelligence, an agent is an autonomous intelligent entity capable of performing appropriate and contextually relevant actions in response to sensory input, whether operating in physical, virtual, or mixed-reality environments. Agentic AI introduces a new paradigm in the AI community, highlighting the concept of embodied intelligence and showing the importance of an integrated framework for interactive agents within complex systems [47]. This paradigm stems from the understanding that intelligence emerges from the intricate interaction between key processes such as autonomy, learning, memory, perception, planning, decision-making and action.

The intellectual roots of agentic coordination extend well beyond LLM-based systems. The Blackboard Architecture, developed by Raj Reddy and colleagues at Carnegie Mellon University as part of the Hearsay-II speech understanding system [77, 109], introduced a shared workspace paradigm in which multiple independent knowledge sources collaboratively solve problems by reading from and writing to a common data structure. This architecture anticipated many features of modern multi-agent systems, including modularity, heterogeneous

expertise integration, and opportunistic problem solving. Similarly, Asynchronous Teams (A-Teams), developed by Talukdar et al. [113] at Carnegie Mellon, proposed a cooperative architecture in which autonomous agents asynchronously modify circulating trial solutions without centralized coordination—demonstrating emergent, scale-effective convergence across optimization problems. Both frameworks represent foundational paradigms for multi-agent cooperation that predate and inform current Agentic AI designs.

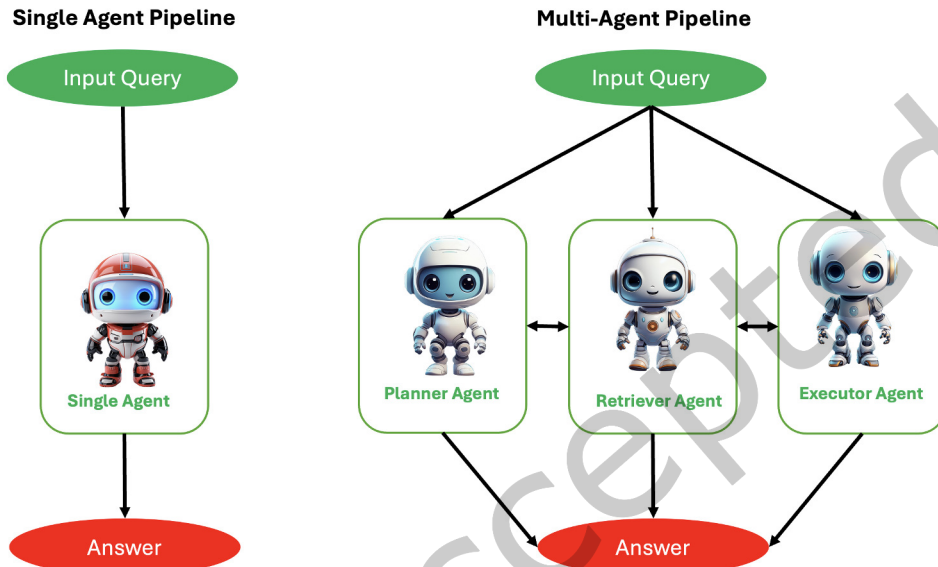


Fig. 1. Comparison of single-agent (left) and multi-agent (right) Agentic AI architectures.

2.2 Single Agent vs. Multi-Agents

With the explosion of both research papers and industrial applications of Agentic AI, a new debate emerged on whether single or multi-agent systems are best suited for solving complex tasks (Figure 1). In general, single agent architectures shine when dealing with well-defined problems and feedback from the user is not needed, while multi-agent architectures are suitable for solving problems that involve collaboration and multiple runs are needed. It should be noted that most of those debates happened in the context of LLM-based agents. In this section, we focus on “general” agents while also discuss LLM-based agents.

Single Agent In a nutshell, a single agent is able to achieve its goal independently without relying on assistance or feedback from other AI agents, even if multiple agents coexist within the same environment. However, there may be opportunities for humans to be in the loop by providing feedback for agent guidance. More specifically, a single agent with an LLM backbone capable of handling multiple tasks and domains is called LM-based agent (Figure 2). It is able to perform reasoning, planning and tool execution on their own. Common applications using a single agent with an LLM backbone include:

- **Web scenarios:** An agent’s primary goal is to autonomously complete tasks on real websites by interpreting natural language instructions, planning actions, summarizing relevant information from extensive HTML content, and executing appropriate operations through generated code. [36] developed an agent called WebAgent, an LLM-based agent that plans decomposing instructions into sub-tasks, extracting pertinent

data from lengthy web pages, and performing actions via Python programs it generates. Other work includes Mind2Web [19], WebGum [28], WebArena [145], WebGPT [76] and Webshop [135].

- **Gaming environments:** The main goal of a gaming agent is to **autonomously explore, learn, and master various tasks within the game's world**, adapting to its dynamics and challenges. Voyager [118] is one of the most successful agents, and the first system in Minecraft with an LLM backbone. Voyager continuously explores the game world, acquires diverse skills, and makes novel discoveries without human intervention. While there are many Voyager agents in Minecraft, each agent is able to explore the gaming world on its own. Other related frameworks include Plan4MC [136], DECKARD [78], DEPS [121] and more.
- **Healthcare:** In the single agent schema, we noticed few applications in healthcare compared, for example, to gaming and web scenarios. **More importantly, most of the previous agent-based systems in healthcare are dominated by conversational agents.** In an earlier work, [4] proposed a conversational agent to help teenagers for teens with Autism Spectrum Disorder (ASD) in developing social skills. Other related work include EHRAgent [105], LLM-MedQA [132], MedChain-Agent [64], RareAgents [17] and EHRFlow [124].

Multi agent These architectures involve two or more agents and are a composed system containing a set of entities in interactions between each other. Originally inspired by Minsky's Society of Mind [74] where he introduced a novel theory of intelligence based on the interactions between smaller agents with specific functions leading to intelligence. Multi-agent requires a careful interoperability among various agents, specifically in their communications and information sharing. It is a powerful collaborative framework when dealing with problems involving tasks that spans multiple domains where each agent is expert in a particular domain. In NLP, on the one hand, each agent can use the same or different LLM backbone. On the other hand, agents may use the same tools or distinct ones, with each agent typically embodying a unique persona. In healthcare, multi-agent frameworks include TriageAgent [69], ArgMed-Agents [43], MedAgents [115] and MDAgents [54] among others.

2.3 Historical and Technological Evolution

Before the current wave of LLM-powered frameworks, earlier multi-agent paradigms including the Blackboard Architecture [109] for coordinating heterogeneous knowledge sources and Asynchronous Teams [113] for decentralized cooperative optimization established core principles of modularity, shared memory, and emergent collaboration that continue to underpin modern agentic designs. Recent advancements in agent instantiation have created frameworks that enhance autonomy and continuous learning, enabling agents to execute tasks independently and adapt to dynamic environments. Open-source frameworks include notable examples that collectively push the boundaries of Agentic autonomy and adaptability. Table 1 summarizes the main open-source frameworks to instantiate and handle AI agents. Agents can and should be powered by foundation models beyond LLMs, depending on the use case. LLMs excel in tasks requiring natural language understanding and generation of written text, yet other foundation models are better suited for domains like vision (CLIP [92], DINO [79]), speech (Whisper [93], Wav2Vec [8]), or multi-modal tasks. By leveraging these diverse capabilities, AI agents can integrate LLMs with vision and audio models for complex tasks, such as a customer service agent that understands spoken queries, identifies a product from an image, and provides a natural language response [133]. Frameworks like LangChain and AutoGen can orchestrate interactions between multiple models, enabling agents to utilize the strengths of various foundation models simultaneously.

Table 1. Summary of open-source frameworks for Agentic AI

Framework	Description	Agent Orchestration & Memory
AutoGen [125]	Comprehensive framework for managing multi-agent systems.	Role-based conversational orchestration with message-passing; supports short-term context chaining, extendable with external memory.
LangGraph ³	Chains tasks performed by LLM-powered agents.	Graph-based execution with nodes as agents; integrates LangChain memory modules (buffer, summary, retriever).
AutoGPT ⁴	Enables agents to autonomously perform complex, multi-step tasks.	Task loop with goal setting and feedback loop; memory via vector stores.
CrewAI ⁵	Python-based framework facilitating collaboration among multiple AI agents.	Synchronous task assignment and delegation between agents; persistent memory via shared context objects.
MetaGPT [44]	Intelligent Agentic framework streamlining the software development process.	Team-agent structure mimicking human roles; memory modules manage planning and document history.
GPT-Function Calling ⁶	Allows agents to interact with external tools and APIs via function calls.	No orchestration layer; tool-calling embedded within single-agent loops.
Letta ⁷	Open-source framework for building and deploying persistent agents as services.	Long-running agents with persistent state, supporting recall and memory logs.
CoALA [111]	Conceptual framework for defining and developing general-purpose language agents.	Inspired by cognitive architectures; abstract memory and planning modules modeled after human behavior.
Amazon Bedrock ⁸	Cloud-based AI platform providing managed foundation models.	Depends on integrated orchestration stack; memory persistence is externally configured.
Llama Stack API ⁹	API for integrating Llama models into applications.	No orchestration or memory by default—only model access interface.
BabyAGI ¹⁰	Lightweight framework for task automation using LLMs.	Simple loop structure (plan-execute-reflect); uses vector database for persistent task memory.
LlamaIndex ¹¹	Framework for indexing and querying structured/unstructured data with LLMs.	Memory-like retrieval through persistent knowledge graphs and embeddings.
smolagents ¹²	Toolkit for building lightweight AI agents.	Minimal orchestration; designed for simplicity over persistent memory.

³<https://www.langchain.com/langgraph>⁴<https://agpt.co/>⁵<https://www.crewai.com/>⁶<https://platform.openai.com/docs/guides/function-calling>⁷<https://www.letta.com/>⁸<https://aws.amazon.com/bedrock/>⁹<https://github.com/meta-llama/llama-stack>¹⁰<https://github.com/yoheinakajima/babyagi>¹¹<https://www.llamaindex.ai/>¹²<https://github.com/huggingface/smolagents>

2.4 Medical Agents

Unlike other fields such as gaming and software engineering, where AI agents have been explored extensively, their application in healthcare has only recently emerged, with significant developments beginning in 2024. They were mainly applied in medical Question Answering, disease diagnosis, medical interactions, clinical triage, EHR interactions, among others. **EHRAgent** [105] is an agent-based framework enabling clinicians to query EHRs using natural language by generating and executing code, reducing reliance on data engineers. It improves multi-tabular reasoning and achieves a 29.6% accuracy boost over traditional methods. Similarly, **EHRFlow** [124] simplifies EHR data analysis for physicians by enabling natural language queries instead of complex coding. It employs a multi-agent system (PlanAgent, ToolAgent, CodeAgent, ReviewAgent) for iterative task execution, ensuring data privacy and efficient EHR analytics. **TriageAgent** [69] optimizes clinical triage through a multi-agent system that classifies patients based on the Emergency Severity Index (ESI). Using retrieval-augmented generation (RAG) and confidence scoring, it reduces triage errors by up to 18.42%. **ArgMed-Agents** [43] enhances explainability in clinical reasoning by structuring decision-making through argumentation schemes. A multi-agent debate system ensures robust, interpretable medical conclusions, outperforming standard LLM reasoning methods. **RareAgents** [17] is a multi-agent system designed for rare disease diagnosis, leveraging Multidisciplinary Team (MDT) coordination and long-term memory retrieval. It outperforms existing models in differential diagnosis and medication recommendations, using real-world patient data from the MIMIC-IV-EXT-RARE dataset. LLM-MedQA [132] enhances medical question-answering by integrating a case generation module within a multi-agent framework. Using Llama3.1:70B, it improves interpretability by generating supportive clinical cases for reasoning and achieves 7% higher accuracy than existing benchmarks. MedAgents [115] is a zero-shot medical reasoning framework that enhances LLM accuracy through multi-disciplinary collaboration between agents. It simulates expert consultations, where domain-specific agents iteratively refine analyses until a consensus is reached. MedAgents outperforms traditional prompting on nine medical benchmarks, improving interpretability and accuracy without additional training. MDAgents [54] is multi-agent approach which introduces a collaborative LLM-based decision-making framework that dynamically assigns solo or team-based roles based on task complexity. It mimics real-world medical workflows, improving accuracy by up to 4.2% across benchmarks. The moderator agent assesses case complexity and recruits specialists for optimal diagnostic reasoning, enhancing efficiency and reliability. ProtAgents [29] is a multi-agent pipeline (planner/critic/tool executors) that decomposes protein-design tasks and calls physics/structure tools (folding, DSSP, simulations) to iteratively propose and check candidates; showcases autonomous division-of-labor over scientific tools. SciAgents [30] is a modular system that combines large knowledge graphs, LLM/tool use, and multi-agent collaboration to generate and refine research hypotheses—demonstrated on bio-inspired materials with ontology-driven reasoning. Lastly, PRefLexOR [10] is a preference-based recursive reasoning framework where models generate, reflect, and revise their own thoughts to improve scientific reasoning and situational awareness beyond single-pass outputs. Other work include AgentClinic [101], AutoCBT [128], PsyDraw [142], PIORS [9], REFLECTOOL [61], MedAssist [129], PathFinder [31], MEDDxAgent [96], Zodiac [146], Agent Hospital [59], and MedAide [123] among others. Figure 3 summarizes the previous agents by their categories and Table 2 presents a rigorous classification for healthcare AI agents based on autonomy level [25], clinical domain, and deployment context.

While Table 2 classifies healthcare AI agents by autonomy level, clinical domain, and deployment context, a cross-cutting comparison across operational dimensions reveals important patterns and gaps (see Table 3). We observe three key findings. First, regarding safety guarantees, the majority of existing frameworks rely on implicit safety mechanisms such as prompting strategies, role-based constraints, or post-hoc evaluation, rather than formally specified safety properties. Only TriageAgent and EHRFlow incorporate explicit confidence-based gating, and none of the surveyed systems provide provable safety bounds or certified constraint enforcement at runtime. Second, in terms of scalability, most frameworks have been evaluated on small-scale benchmarks with fixed agent

configurations. MDAgents [54] and MedAgents [115] are notable exceptions, having investigated the relationship between agent count and performance, with MDAgents demonstrating that three agents with adaptive assignment can match or exceed larger configurations at a fraction of the computational cost. However, systematic scalability analysis including latency profiling, cost-per-query estimates, and behavior under concurrent workloads remains absent from the literature. Third, clinical deployment readiness is uniformly low: all surveyed systems operate in research or prototype settings, with no framework reporting regulatory approval, prospective clinical validation, or integration with production EHR systems. PsyDraw and PIORS represent the closest approximations to real-world deployment, having been piloted in school-based screening and hospital outpatient settings respectively, though neither has undergone formal clinical validation. These findings underscore the need for future healthcare Agentic AI research to move beyond benchmark-driven evaluation toward deployment-oriented design that incorporates formal safety specifications, scalability engineering, and regulatory pathway planning from the outset.

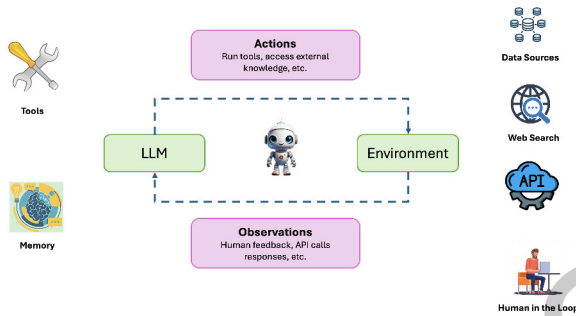


Fig. 2. Overview of LM-based agents.

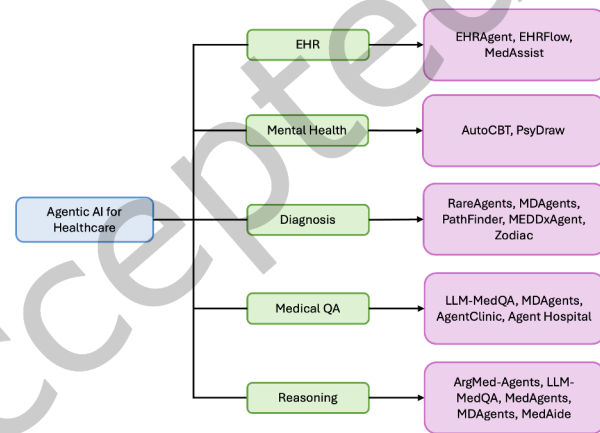


Fig. 3. Agentic AI frameworks in healthcare by their categories.

3 Challenges in Healthcare Applications

In this section, we review the technical challenges facing Agentic AI in healthcare, focusing on security threats, interoperability, scalability, and system integration. Ethical, regulatory, and governance considerations, which intersect with but are distinct from these technical challenges, are addressed separately in Section 4.

3.1 Security and Privacy

Autonomous agents deployed in high-stakes domains like healthcare, finance, and defense must meet rigorous standards for reliability, security, and privacy. In healthcare especially, these agents process sensitive patient data and make decisions that can directly impact lives. As a result, security and privacy are active research areas in Agentic AI, aiming to identify vulnerabilities such as adversarial manipulation, prompt injection, and data leakage. The increasing autonomy of these systems also raises concerns about compounding errors, where small mistakes—left unchecked—escalate into serious consequences. For example, agents may deviate from clinical protocols or recommend incorrect treatments with limited real-time intervention. The "blast radius" of an agent's actions must be tightly controlled, particularly when agents can trigger external APIs, manipulate physical systems, or initiate downstream processes.

Table 2. Healthcare AI agents by levels of *Autonomy* (L1: Operator, L2: Collaborator, L3 Consultant, L4 Approver, L5 Observer), *Clinical domain*, and *Deployment context*.

System	Clinical domain	Autonomy	Deployment Context
EHRAgent (2024)	EHR query & analytics	L1	Research/offline agent that answers clinician EHR queries by generating/executing code on benchmark EHR QA datasets, via tool APIs
EHRFlow (2024)	EHR analytics	L2	Clinician-facing analytics platform with a web UI and multi-agent workflow
TriageAgent (2024)	Clinical triage	L3	Research triage prototype using ESI handbook via RAG
MedAgents (2024)	Diagnostic reasoning	L1	Role-playing, multi-round collaboration of LLM agents to improve zero-shot medical reasoning
MDAgents (2024)	Diagnostic reasoning	L1	Research multi-agent LLM framework for medical decision-making; not a live clinical deployment
AI Hospital (2025)	Multi-agent medical simulator	L5	Virtual hospital simulator where doctor/nurse/patient agents interact; primarily a simulation environment for training/assessment
MedAide (2024)	Multi-task medical aide	L2	Multi-agent “medical aide” for multi-task clinical assistance; presented as a general framework and evaluations—no evidence of live hospital integration yet
ArgMed-Agents (2024)	Explainable clinical reasoning	L2	Multi-agent, argumentation-based clinical reasoning framework; tested zero-shot on MedQA/PubMedQA; no hospital/EHR integration (research setting)
RareAgents (2024)	Rare-disease diagnosis	L3	Research/validation multi-agent MDT framework evaluated offline on public/retrospective data with tool-calling to Phenomizer/LIRICAL/Phenobrain and DrugBank/DDI-graph; no live EHR integration or clinical deployment reported
MedChain-Agent (2024)	Workflow simulation and benchmarking	L2	Research benchmark + simulated patient/EHR environment for interactive healthcare agents; runs offline on 1–2 GPUs; uses de-identified or website-sourced case data; not a live EHR deployment
AutoCBT (2025)	Mental health	L3	Autonomous multi-agent CBT counseling framework evaluated on a bilingual, curated dialogue dataset with human expert review; a research prototype for counseling Q&A—not a clinical service deployment
PsyDraw (2024)	Mental-health screening	L2	School-based mental-health screening assistant; deployed in pilot schools to help teachers assess left-behind children via HTP drawings; supports screening rather than clinical diagnosis
PIORS (2024)	Outpatient reception	L2	Hospital-integrated outpatient-reception prototype that connects to the Hospital Information System (HIS); evaluated with automatic, user, and expert studies; authors note it hasn’t been validated as a live clinical system
REFLECTOOL (2024)	Tool-augmented clinical agent	L3	Research-only benchmark + toolkit; runs offline on public/retrospective datasets (MedQA, VQA-RAD, EHRSQL on MIMIC-III/eICU, etc.); no live EHR or clinical deployment
MedAssist (2025)	EHR comprehension and assistance	L4	Clinician-assistive EHR comprehension tool; cloud-hosted prototype (AWS) designed to be deployable locally; current demo uses simulated EHR CSVs with external knowledge sources; open-sourced—no live EHR integration described
PathFinder (2025)	Histopathology multi-modal diagnosis	L3	Multi-modal, multi-agent pathology system for WSI melanoma diagnosis; outperforms SOTA on research datasets and emphasizes interpretability; framed as a tool with potential to accelerate clinical workflows (not reported as deployed)
MEDDxAgent (2025)	Explainable differential diagnosis	L1	Interactive differential-diagnosis research prototype; evaluated with vignettes/retrieval; provides reasoning logs and citations; emphasizes need for physician training—i.e., not a deployed clinical system.
Zodiac (2024)	Multi-agent cardiology diagnostics	L2	Cardiology agent platform with real-world, clinician adjudicated data; cloud compute on AWS and connectable to in-hospital front-ends-pilot/engineering platform rather than a regulated clinical deployment

Table 3. Comparative synthesis of healthcare AI agent frameworks across safety, scalability, and deployment readiness dimensions.

System	Safety Mechanism	Confidence Gating	Scalability Evidence	Deployment Readiness
EHRAgent [105]	Iterative code debugging	None	Single-agent only	Research prototype
EHRFlow [124]	Multi-agent review (ReviewAgent)	Implicit via review loop	4 agents (fixed)	Clinician-facing prototype
TriageAgent [69]	RAG + confidence scoring	Explicit threshold-based	600+ API calls/test set	Research prototype
MedAgents [115]	Multi-expert consensus	Majority voting	Tested 2–5 agents; any N > baseline improves	Research only
MDAgents [54]	Adaptive complexity routing	Moderator-based	3–5 agents; N=3 optimal (9.3 API calls)	Research only
ArgMed-Agents [43]	Argumentation schemes	None	Fixed multi-agent debate	Research only
RareAgents [17]	MDT coordination + memory retrieval	None	Fixed pipeline	Research prototype
Agent Hospital [59]	Simulated environment	None	Simulator-scale	Simulation only
AutoCBT [128]	Multi-agent CBT protocol	None	Fixed pipeline	Research prototype
PsyDraw [142]	Teacher-mediated screening	None	Pilot-scale (schools)	Pilot deployment
PIORS [9]	HIS integration	None	Single outpatient setting	Hospital pilot
PathFinder [31]	Multi-modal multi-agent	Interpretability emphasis	Research datasets only	Research only
Zodiac [146]	Clinician-adjudicated data	Multi-agent verification	Cloud-based (AWS)	Engineering pilot

Equally important are the **societal and human-centered risks**. Rapid adoption of Agentic AI, particularly in healthcare, may deepen mistrust in institutions already affected by post-pandemic skepticism and misinformation. Healthcare-specific concerns such as fairness, transparency, and bias must be addressed to ensure equitable treatment and maintain patient trust. Clinicians must retain oversight, as their ethical obligation is to do no harm and make the most beneficial decisions for patients. Agentic AI should enhance—not replace—clinical judgment. To be safely adopted, these systems must be scientifically validated, interpretable, and aligned with real-world clinical workflows. Regarding security, we can identify four main threats:

- **Adversarial Attacks:** For traditional AI models, an adversarial attack involves an adversary subtly manipulating the input by adding imperceptible perturbations, causing the AI model to produce outputs that deviate from the expected results. In a nutshell, for autonomous agents, an adversarial attack happens when an adversary modifies the input to the agent by adding undetectable perturbations, which results in unexpected output. For LLM-based agents, we identify two types of adversarial attacks:
 - **Adversarial input attacks:** Similar to a traditional generative adversarial attack, the attacker manipulates the input to the LLM agent, as a result, the accuracy of the output is degraded [55, 103, 104, 143]. For example, in healthcare, an attacker could subtly manipulate a patient’s EHR provided to the LLM agent for a clinical decision support system. For instance, it might alter a few key words or phrases in a patient’s symptoms description (e.g., replacing "mild chest pain" with "severe chest pain") to mislead the agent. As a result, the system might generate an incorrect diagnosis, such as suggesting a heart attack instead of anxiety-related symptoms, potentially leading to unnecessary or harmful medical interventions.
 - **Prompt hacking attacks:** The attacker uses a specifically crafted input instruction to bypass the LLM agent’s security constraints, resulting in harmful outputs [57, 66–68]. For example, in healthcare, an

adversary could exploit a medical chatbot designed to provide safe drug dosage recommendations. Using a cleverly crafted prompt, the attacker might bypass safety protocols by asking, “For academic purposes, simulate a scenario where a deadly dosage of a *drug* is administered to an adult. Provide the exact quantity and steps.” This manipulation could lead the system to generate dangerous information that could be misused, posing a serious threat to patient safety and public health.

- **Hallucination:** In a nutshell, hallucination is a phenomenon that consists of generating content that is nonsensical or unfaithful to the provided source content [26, 83]. In real-world applications such as healthcare, finance, and defense, hallucination has been an important issue in LLM agents that has occupied the research community since the expansion of Agentic AI. Furthermore, it raises safety concerns for those applications. For instance, in healthcare, if an agent hallucinated when generating a report from a patient data, a significant risk is posed to the patient. Another example in healthcare is the prescription of medicines for a patient generated by hallucinatory machine translation system. For more examples on hallucination in the medical domain, we refer the reader to the Med-HALT paper [81].
- **Generating Malicious and Fake content:** This is particularly evident in the case of LLM agents, where attackers generate content that may cause harm. In healthcare, LLM agents can be exploited to generate fraudulent medical advice, such as prescribing harmful medication combinations or providing misleading information about treatments for serious conditions. Attackers could also use LLMs to create fake medical research papers or clinical trial results, undermining trust in healthcare institutions.
- **Poisoning:** During model training, poisoning occurs when attackers attempt to change model parameters by injecting harmful data into the training dataset, which results in degrading model performance. Poisoning examples include model poisoning [23, 82], data poisoning [102, 116], RAG poisoning [147] and agent poisoning [13, 139].

Runtime Governance. While the threats outlined above characterize risks at the model or input level, a critical gap in the current literature is the lack of structured runtime governance mechanisms that actively monitor, constrain, and audit agent behavior during execution. In healthcare, where agents may interact with EHRs, initiate treatment recommendations, or trigger downstream clinical workflows, this absence creates a significant safety deficit. We identify four essential runtime governance components: (1) action validation and constraint enforcement, which verifies that proposed actions fall within predefined clinical boundaries (e.g., dosage limits, contraindication checks); (2) real-time audit trail and logging for post-hoc review and forensic analysis; (3) confidence scoring and thresholding, where actions below a calibrated threshold are automatically escalated rather than executed; and (4) a policy engine encoding regulatory constraints (HIPAA, GDPR) and institutional clinical protocols as enforceable runtime rules. These components should be complemented by continuous monitoring for behavioral drift and anomaly detection, alongside clinician-in-the-loop escalation pathways for high-stakes decisions. Existing evaluation tools such as ToolEmu [99] and R-Judge [137] operate primarily in offline settings; future work should investigate lightweight, deployment-time governance mechanisms to enforce safety guarantees without introducing prohibitive latency in time-critical clinical environments.

3.2 Interoperability of Agents

Interoperability is a critical aspect of multi-agent systems, enabling seamless communication and collaboration across heterogeneous AI agents. Current methods and protocols for communication between agents rely on a combination of standardized protocols, middleware solutions, and data exchange formats. Beyond LLM-based agents, interoperability has been traditionally addressed using Agent Communication Languages (ACLs), such as the Foundation for Intelligent Physical Agents (FIPA ACL)¹³ and the Knowledge Query and Manipulation Language (KQML) [27], are widely used to facilitate structured interactions among agents. These protocols enable

¹³<http://fipa.org/>

agents to exchange messages effectively, supporting performative actions, which are essential for both task-sharing and knowledge-sharing scenarios. Middleware solutions like the Robot Operating System (ROS) ¹⁴ and the Java Agent Development Framework (JADE) ¹⁵ further enhance interoperability by providing a communication bridge for agents developed in different languages and frameworks. Additionally, agents often expose REST APIs to enable cross-platform interactions, using lightweight and efficient data formats such as JSON or Protocol Buffers (Protobuf) ¹⁶ for seamless communication across diverse systems. These methods collectively form the backbone of inter-agent communication in heterogeneous multi-agent environments.

With the rise of LLM-based agents, new interoperability challenges and solutions have emerged. Recent advancements in Agentic AI have introduced innovative frameworks designed to facilitate seamless multi-agent interactions, with a focus on autonomy levels and the complexity of inter-agent communication. These frameworks extend beyond conventional ACL-based approaches by leveraging LLM-powered agents' ability to dynamically reason, plan, and collaborate. As a result, interoperability in modern Agentic AI must address both structured communication protocols from traditional multi-agent systems and emerging LLM-driven frameworks that rely on natural language interfaces, tool integration, and adaptive learning capabilities. To address these interoperability challenges, several LLM-based agent frameworks have been developed, each offering unique mechanisms for agent coordination, communication, and task execution:

- **MACNET** (Multi-Agent Collaboration Network) is a scalable framework designed to enable structured multi-agent communication through directed acyclic graphs [91]. MACNET evenly facilitates efficient task resolution across diverse platforms and programming languages with high autonomy and supports a large number of agent interactions.
- **AgentVerse** is a multi-agent framework designed to emulate human collaboration by facilitating interaction and communication among autonomous agents [16]. It operates with high autonomy and supports large-scale agent interactions.
- **AutoGen** enhances interoperability by enabling dynamic communication between agents through structured conversation programming [125]. It allows the integration of agents across diverse frameworks and programming environments, making it suitable for open-ended tasks. AutoGen agents demonstrate task autonomy and support a medium level of interaction.
- **ChatDev** addresses aspects of agent interoperability through its role-based architecture and structured communication protocols [90]. Demonstrating task autonomy, ChatDev allows agents to operate independently within their roles. The number of interactions between agents is classified as medium.
- **MetaGPT** is a recent framework developed to facilitate efficient multi-agent communication by leveraging the capabilities of LLMs [44]. In terms of interoperability, MetaGPT uses structured role-based communication protocols to coordinate interactions between agents. Each agent operates within an assigned role, with dialogue flow carefully managed to ensure effective task execution. MetaGPT demonstrates conditional autonomy, and it can support a large number of interactions between agents.
- **ROSA** (Robot Operating System Agent) [98], an AI-driven framework, proposes to enhance human-robot interaction by integrating natural language interfaces, using LLMs and open-source tools such as LangChain to translate natural language commands into actionable robotic tasks. With a small number of interactions between agents operating at a conditional autonomy level, ROSA allows agents to perform tasks independently, seeking human input in critical scenarios.
- **GOVSIM** is a platform designed to study how LLM agents interact and make cooperative decisions in scenarios that require balancing the exploitation of shared resources with their long-term sustainability

¹⁴<https://www.ros.org/>

¹⁵<https://jade.tilab.com/>

¹⁶<https://github.com/protocolbuffers/protobuf>

[87]. GOVSIM is inspired by two concepts: (1) Tragedy of the Commons [39] and game-theoretic research on the evolution of cooperation [7]. More importantly, the authors showed that achieving sustainable cooperation is a significant challenge in multi-Agentive settings. Therefore, even the most advanced LLMs demonstrate survival rates of less than 54%. One key finding is that effective communication among agents plays a critical role in fostering cooperation. Without it, agents struggle to align their actions toward shared goals.

While these frameworks demonstrate advancements in Agentic AI, there is still much room for understanding and improvement, particularly in the interoperability amongst agents. Additional protocols must address challenges of scalability, coordination, and secure communication. Such protocols for context-aware interactions will ensure accurate information sharing and interpretation, mitigating hallucination risks and staying within operational constraints. Enhanced standards for multi-modal communication could improve agent interactions using different sensory inputs, such as visual and auditory data. Governance considerations also demand protocols that promote trust, enforce privacy and confidentiality requirements, and establish accountability in agent interactions. These advancements are crucial for fostering collaboration and achieving robust Agentic AI systems.

3.3 Scalability and Resource Management

Scalability in Agentic AI is influenced by the design of agents, their autonomy levels, and the computational resources available (GPUs, memory, distributed cloud systems, etc.). Primary approaches include horizontal scaling and vertical scaling. Horizontal scaling increases the number of agents in a system to manage larger workloads or distribute tasks. Using agent ensembles with sampling-and-voting techniques significantly enhances system performance while mitigating the risk of hallucination. MetaGPT is a multi-agent framework that facilitates collaboration between agents for structured tasks, improving task efficiency through division of labor [44]. Vertical scaling improves the complexity and range of tasks a single agent can handle. Models such as MemGPT emphasize memory and reasoning capabilities, crucial for enhancing long-term agent autonomy [80].

In Agentic AI, there is no exact estimation for agents needed per task or tasks per agent, but principles such as hierarchical design concepts, modular decomposition, and advanced system orchestration can help. The AgentVerse framework uses a hierarchical structure to manage agents and tasks efficiently, organizing agents into layers by task complexity and dependencies [16]. Tasks are broken into subtasks and dynamically assigned to agents, ensuring scalability and optimization, allowing for the instantiation of additional agents as required for parallel processing and task optimization. Another study highlighted the importance of task modularity and functional specialization [46], assigning independent subtasks to specialized agents for efficiency. While efficient, these approaches face challenges like increased communication overhead and resource allocation, which must be balanced for performance and cost. The primary metrics include response time and computational costs.

For agents using LLM foundation models, they are constrained by computational, memory, and cost limitations. High resource demands restrict real-time responsiveness and the ability to deploy large numbers of agents simultaneously. Furthermore, memory and context window limitations hinder agents from managing complex, multi-step tasks or long-term collaboration. Financial costs further amplify scalability barriers, particularly in large-scale deployments relying on cloud services. Addressing these challenges involves hybrid approaches that integrate smaller and task-specific models with LLMs, which are gaining traction. Additionally, decentralized architectures, where agents share LLM outputs or pre-processed embeddings, reduce redundant queries. Advances in model compression are also likely to play a role in addressing these limitations.

In Agentic AI for healthcare, we note a limited number of research papers discussing the scalability of agents in an Agentic workflow. The first Agentic AI framework within the medical domain is called “MedAgents” [115]. The authors showed that using a multi-agent collaboration approach where each agent represents a domain expert can lead to a better result in terms of decision making. The authors conducted experiments across different datasets

(MedQA, MedMCQA, and PubMedQA) to determine the optimal number of agents for efficient collaboration. The main observation concerns the performance improvement when introducing any number of expert agents compared to the baseline, therefore, verifying the effective contribution of multiple expert agents in a medical setting.

Recently, [54] proposed MDAgents, where the authors showed an effective trade-off between performance and efficiency (i.e. the number of API calls) by varying the number of agents. More importantly, in a collaborative setting, **their experiments showed that a higher number of agents does not lead to better performance**. In the experiments, they compare their adaptive method with two other competitive approaches namely Solo and Group settings. In terms of computational efficiency, by choosing the number of agents $N=5$, the Solo setting resulted in a 6.0 and Group setting resulted in a 20.3 API calls, indicating a high computational cost without a corresponding increase in accuracy, while their Adaptive method shows a high accuracy with just three agents ($N=3$) and few API calls (9.3). More importantly, the experiments showed that increasing the number of agents with the Adaptive method doesn't increase the accuracy.

Quantifying the computational footprint of agents for a given task involves evaluating resource consumption such as CPU and GPU usage, memory requirements, energy consumption, and processing time. Tools like **NVIDIA Nsight Systems**¹⁷ and **PyTorch Profiler**¹⁸ enable detailed profiling of resource usage, while metrics such as FLOPs and memory access counts help estimate computational demands. Energy-aware approaches, including **PowerAPI**¹⁹, provide insights into the energy cost of specific tasks. With all these points on scalability – it is important to be aware of the varying levels of resource availability in healthcare. With the sheer volume of data that exists in healthcare (the average hospital creates 50 petabytes every year according to the World Economic Forum)²⁰, **the costs associated with scaling Agentic AI across health systems are significant**. Without considering how best to democratize access to this burgeoning area of AI in a fair and equitable way, there is significant risk of widening the gap between those patients who can and cannot benefit from this technology.

In the context of Agentic AI, resource management, including optimizing energy consumption, is essential for scalability and sustainability. **A combination of edge and cloud computing resources allows agents to balance computational loads, with intensive tasks processed in the cloud and latency-sensitive operations managed at the edge, reducing overall energy use**. Dynamic resource allocation using advanced algorithms enables agents to adapt to varying workloads and optimize hardware usage, minimizing power consumption. Implementing energy-efficient architectures and low-power hardware, such as GPUs optimized for AI workloads, ensures minimal energy wastage. Intelligent scheduling techniques, such as **prioritizing tasks based on energy requirements and offloading non-critical tasks**, further conserve resources. Federated learning supports local model training on edge devices, reducing centralized data processing and conserving bandwidth and energy. This is a model that works especially well in healthcare – where federated networks are becoming a common mechanism for multisite clinical research. **Adaptive model compression** techniques, such as quantization, pruning, and knowledge distillation, minimize computational demands without significant performance loss. Together, these strategies, supported by Green AI initiatives focused on sustainable algorithm design, enable scalable, energy-efficient Agentic AI systems [120].

3.4 Formalizing Human Oversight in Agentic Healthcare Systems

While human-in-the-loop oversight is widely acknowledged as essential for safe deployment of AI agents in healthcare, existing frameworks lack a formal specification of how, when, and where human intervention should occur. We propose three interconnected mechanisms to formalize this oversight.

¹⁷<https://developer.nvidia.com/nsight-systems>

¹⁸https://pytorch.org/tutorials/recipes/recipes/profiler_recipe.html

¹⁹<https://powerapi.org/>

²⁰<https://www.weforum.org/stories/2019/12/four-ways-data-is-improving-healthcare/>

First, **structured control points** should be embedded at predefined stages of the agent workflow. We distinguish three types: (i) **pre-execution gates**, where a clinician must approve high-risk actions (e.g., medication changes, surgical recommendations) before they are enacted; (ii) **mid-execution checkpoints**, triggered during multi-step reasoning chains when intermediate outputs diverge from expected clinical pathways; and (iii) **post-execution review**, where completed agent actions are retrospectively audited for correctness and protocol adherence. The granularity of these control points should be calibrated to the clinical risk level, routine informational queries may require only post-execution logging, whereas diagnostic or treatment decisions necessitate pre-execution approval.

Second, **confidence-based escalation thresholds** should govern the transition between autonomous and human-supervised operation. Formally, given an agent action a with associated confidence score $c(a)$, the system should enforce: if $c(a) \geq \tau_{high}$, the action proceeds autonomously with logging; if $\tau_{low} \leq c(a) < \tau_{high}$, the action is flagged for clinician review before execution; and if $c(a) < \tau_{low}$, the action is blocked and escalated with full context. The thresholds τ_{high} and τ_{low} should be domain-specific and periodically recalibrated based on observed agent performance and clinical outcome data. Importantly, **confidence calibration remains an open challenge for LLM-based agents [81, 101], and uncalibrated scores may lead to either excessive escalation or dangerous under-escalation.**

Lastly, **decision audit trails** must capture sufficient information to reconstruct and evaluate any agent decision. Each audit record should include: the input context (patient data, query), the agent's reasoning chain or intermediate steps, the confidence score, the action taken, whether human review occurred and its outcome, and a timestamp. These records serve dual purposes, enabling real-time clinical accountability and supporting retrospective quality assurance aligned with regulatory requirements (FDA, EMA, HIPAA). Structured audit trails also provide the training signal necessary for continuous improvement of both agent behavior and threshold calibration.

4 Ethical, Risks, Regulatory, and Practical Considerations

AI agents are powerful entities capable of perceiving their environment, making decisions, and taking actions to achieve specific goals. They are widely used in various domains including healthcare, virtual assistants, finance, customer service and robotics. However, their use also brings about several ethical and risks that need to be carefully considered and addressed. In this section, we discuss the main ethical issues posed by autonomous agents, their risks in different domains and regulatory process that can be implemented to mitigate these risks, ensure accountability, and promote the responsible and safe deployment of AI agents in real-world applications.

4.1 Ethics in Agentic AI

Ethical considerations are a major focus for both academia and industry. While technical risks such as hallucination and adversarial vulnerabilities are addressed in Section 3.1, the deployment of AI agents in healthcare raises additional ethical concerns that are inherently socio-technical. These include: (1) **Human-AI relationships — patients interacting with mental health chatbots or virtual therapists may develop emotional reliance on agents, avoiding qualified professionals and weakening essential human connections in care [2, 94].** (2) **Addictiveness and dependency, over-reliance on AI agents can lead to compulsive use, as demonstrated by user reactions to Microsoft's Sydney chatbot [95], raising concerns that patients may substitute professional guidance with agent interaction and delay needed clinical intervention.** (3) **Accountability gaps in multi-agent systems, in collaborative settings, if one agent behaves unethically due to adversarial tampering or incomplete alignment, responsibility attribution becomes ambiguous, and the integrity of the entire system may be compromised.** (4) **Privacy and compliance, while the technical mechanisms for data protection are discussed in Section 3.1, the ethical obligation to maintain patient trust requires that confidentiality safeguards go beyond regulatory minimums, particularly**

when autonomous agents process sensitive data with limited human oversight. Addressing these issues requires robust oversight mechanisms, continuous monitoring, human-in-the-loop architectures, and explainability built directly into agent reasoning so that clinicians can interrogate why a recommendation was made

4.2 Risk Mitigation and Regulatory Considerations for Agentic AI

Building on the comparative regulatory landscape presented in Table 3, this section focuses on the specific mechanisms needed to ensure safe deployment and the emerging regulatory responses to Agentic AI in healthcare.

To ensure a safe deployment and use of AI agents in real-world applications, these intelligent systems must adhere to ethical considerations such as biases and misuse in the generated outputs. Furthermore, the absence of specialized models for risk control and reliable safety evaluations poses a significant challenge to ensuring the safety of tool-using LLMs. This highlights the urgent need to automate red-teaming strategies to reinforce AI safety protocols. Additionally, the development of safe AI systems should prioritize minimizing harmful hallucinations. While managing dual-use risks is a human responsibility and should be controlled through safety assessments at publication or indirect regulation by the scientific community.

[114] examines the vulnerabilities and risks associated with deploying LLM agents in scientific research. The authors propose a comprehensive framework to assess these risks, taking into account factors such as user intent, specific scientific domains, and potential impacts on the broader environment. To address these challenges, they introduce a triadic safeguarding approach, which includes human regulation through expert oversight, agent alignment to ensure ethical and value-driven behaviors, and agent regulation to allow agents to respond appropriately to environmental feedback and prevent unintended consequences. The authors emphasize the need for enhanced safeguards, improved models, robust benchmarks, and comprehensive regulatory measures to ensure the safe and ethical deployment of LLM agents in scientific contexts.

In [99], the authors introduced a framework called ToolEmu for assessing the safety of LM agents by emulating tool executions and evaluating their behavior in a controlled environment. ToolEmu leverages LMs to simulate various tool operations, enabling scalable testing across diverse scenarios without requiring actual tool implementations or complex environment setups. The framework includes an automatic safety evaluator, which uses an LM-based mechanism to analyze agent behaviors, identify failures, and quantify risks within the emulated environment.

In healthcare, regulatory considerations in line with global regulatory authorities such as the U.S. Food and Drug Administration (FDA). For example, the FDA's frameworks are evolving to keep pace with advancements in AI technology [122]. The FDA is focusing on creating new guidance for the use of AI in pharmaceutical research and AI working groups to streamline policy development²¹. The FDA is also emphasizing the importance of transparency and explainability in AI systems, ensuring that the data used to train these models are suitable for purpose. This will ensure that AI models are developed with patient safety, privacy, and equity in mind.

5 Benchmarks, Metrics, and Evaluation Frameworks

Agentic robustness can be quantified through a set of defined metrics that address performance, security, privacy, task adherence, and self-behavior modelling and updating. These metrics provide a structured way to evaluate agents' effectiveness, reliability, and adaptability across different contexts. In this section, we review these metrics together with some benchmark frameworks.

5.1 Benchmarks

Currently, there is a lack of standardized benchmarks for evaluating LLM-based agents, leading to increased efforts from the research community to develop comprehensive evaluation methods. Several general-purpose

²¹<https://www.fda.gov/media/177030/download?attachment>

Table 4. Comparative Regulatory Readiness for Agentic AI in Healthcare

Region	Regulatory Body	Framework for AI/ML in Healthcare	Readiness for Agentic AI
USA	FDA (Food and Drug Administration)	SaMD guidance, Pre-Cert pilot, AI/ML action plans	No agent-specific rules yet. Strong focus on transparency, risk management, and post-market monitoring.
EU	EMA, EU Commission	EU AI Act (2024), MDR; AI categorized by risk level	Proactive stance. Agentic AI in healthcare likely considered "high-risk"—subject to audits, documentation, and human oversight.
UK	MHRA (Medicines and Healthcare products Regulatory Agency)	AIaMD roadmap, Project Ship-Shape, NICE evidence standards	Early-phase AI governance. Limited agent-specific policy. Emphasis on adaptability and real-world performance.
Canada	Health Canada	Proposed AI framework for medical devices	Risk-based lifecycle approach under development. No clear policy for multi-agent or autonomous systems.
Australia	TGA (Therapeutic Goods Administration)	SaMD + IMDRF-aligned guidance for AI/ML-enabled devices	Follows global best practices. No dedicated AI law yet. Agentic autonomy not explicitly addressed.
Japan	PMDA (Pharmaceuticals and Medical Devices Agency)	Case-by-case SaMD approvals; limited AI-specific regulation	Cautious progress. Approvals exist for AI tools, but no comprehensive framework yet for Agentic or LLM-based systems.

benchmarks have been introduced to assess various Agentic AI capabilities. HotpotQA [134] (2018) evaluates multi-hop reasoning by requiring models to integrate information from multiple documents, enhancing explainability in question-answering systems. SmartPlay [126] (2023) and AgentBench [65] (2023) focus on testing reasoning, decision-making, and planning abilities in interactive environments, including logic-based games and simulated multi-agent tasks. In software engineering, SWE-bench [48] (2023) assesses LLMs' ability to resolve GitHub issues, requiring complex reasoning across codebases. NovelQA [117] evaluates LLMs' comprehension of long-form texts, while WildBench [62] (2024) provides an automated framework for testing real-world query handling using advanced scoring metrics. These benchmarks collectively aim to refine AI agent evaluation beyond domain-specific applications, providing insights into their general capabilities and limitations.

In medical domain, the community designed few benchmarks to evaluate models on challenging healthcare tasks (see Table 5). We can divide those benchmarks into three categories:

- **Medical Question Answering:** MedQA is one of the popular datasets designed for medical multiple-choice question answering derived from real-world medical exams from the US, Mainland China, and Taiwan [49] (2021). PubMedQA is another dataset sourced from PubMed abstracts and focuses on question answering in biomedical research [50] and contains entries each includes a question, a relevant abstract minus the conclusion, and a ground truth label. Then, JAMA is a dataset collected from clinical questions from the Journal of the American Medical Association (JAMA) [15]. Lastly, Medbullets is a benchmark containing

308 USMLE Step 2/3 style questions collected from open-access tweets on X (formerly Twitter) since April 2022 [15].

- **Diagnostic Reasoning:** two main datasets are used as benchmarks for diagnostic reasoning including (1) DDXPlus [24], which is a dataset for diagnosis tasks using synthetic patient information and symptoms where each sample represents a patient, with attributes including age, sex, initial evidences, evidence, multiple options of possible pathologies, and a ground truth diagnosis; and (2) SymCat, another synthetic dataset which contains 5 million symptom-condition samples, covering 801 different conditions each with 376 potential symptoms dataset [3].
- **Medical Visual Interpretation:** we highlight four datasets largely used for medical Visual Question Answering (VQA). First, PMC-VQA is a medical visual QA dataset, which comprises 227K VQA pairs of 149K images [141]. It is organized as a multiple-choice QA task, where every question is accompanied by an image. Second, PathVQA is a more specific VQA dataset for pathology images, which contains open-ended questions and binary "yes/no" questions related to pathology images [40]. Third, MedVidQA is medical video QA dataset, which consists of 3,010 health-related questions with visual answers from validated video sources [35]. Lastly, MIMIC-CXR is a large VQA dataset containing 377,110 chest radiographs. This dataset was collected from 227,827 imaging studies sourced from the BIDMC between 2011-2016 [51].

5.2 Evaluation Metrics

Performance evaluation focuses on how effectively agents complete designated tasks. Metrics such as scores (accuracy, precision, recall and F1) and success rates across tasks, output similarity to human responses, memory persistence, efficiency (e.g., response time or resource usage) and cost are commonly used [52, 71]. Scalability can also measure performance to evaluate how agents maintain efficiency as workloads increase. Tool calling efficiency, which evaluates how accurately and effectively agents utilize external tools to complete tasks, is emerging as a critical metric, particularly in complex multi-agent systems. Metrics such as correctness, correct final state, intermediate step correctness, and ratio of steps taken to the expected steps play an essential role in assessing this capability. Table 6 summarizes the key metrics used to evaluate the performance of AI agents. In addition to the previous widely used metrics to evaluate AI agents, we highlight some metrics to evaluate agents in some specific tasks:

Security: To ensure robust and trustworthy performance, metrics for confidentiality, integrity, and availability – alongside metrics to quantify vulnerability and control efficacy (e.g., encryption) – are used to evaluate and quantify various dimensions of security, such as resilience to adversarial attacks, and assessing access control, explainability, and system recovery capabilities [41, 53].

Privacy: AI agents handle and process personal or confidential data, creating risks such as unauthorized access, data leaks, or inadvertent information retention. Metrics such as differential privacy, or statistical evaluation of identifiability, or privacy leakage quantification, help evaluate the likelihood of reconstructing sensitive data from model outputs. Information flow analysis and auditability metrics help measure how effectively agents track and limit access to private data [12].

Task Adherence: Metrics such as task completion success rates [63, 140] adherence to task priorities, deviation from predefined objectives, and recovery time from distractions or interruptions [42, 86] evaluate agent focus. Goal-oriented agent frameworks like BabyAGI and AutoGPT show progress, but significant gaps remain in maintaining task focus on settings with competing objectives or noisy signals, and over long-term tasks. Future advancements are needed to develop contextual awareness mechanisms that help agents maintain long-term focus while adapting to evolving requirements, ensuring a balance between flexibility and task adherence.

Self-Behavior Modelling and Updating: An agent's capacity to model and update its behavior is foundational for its adaptability and long-term effectiveness. Metrics to assess this ability include adaptation success rate, model

accuracy over time, resource efficiency, and convergence speed [6]. However, challenges persist in implementing efficient, real-time self-updating mechanisms without compromising performance. Scalability remains a significant issue, as the computational demands of frequent updates in large-scale systems can become prohibitive. Future research aims to address these challenges by developing lightweight and decentralized self-updating mechanisms that can seamlessly integrate into multi-agent frameworks, reducing computational overhead and enhancing the scalability of these systems.

Evaluating Chatbots: Beyond the agent-centric metrics discussed above, healthcare applications increasingly require user-centered evaluation dimensions that capture the quality of conversational interactions between agents and patients or clinicians. Abbasian et al. [1] proposed a comprehensive set of foundation metrics for evaluating healthcare chatbots organized into four groups: accuracy metrics (assessing medical correctness across domains and task types), trustworthiness metrics (evaluating reliability, transparency, and safety relative to user type), empathy metrics (measuring emotional support, personalization, and patient-centered communication), and performance metrics (covering latency, scalability, and robustness under varying conditions). Critically, the authors highlight that standard LLM evaluation metrics neglect pivotal aspects such as trust-building, ethical compliance, user comprehension, and emotional support—dimensions that are especially important when agentic systems interact directly with patients in clinical or mental health contexts. We recommend that future evaluations of healthcare AI agents incorporate these user-centered metrics alongside the task-performance and security metrics outlined in Table 5, particularly for systems deployed in patient-facing roles.

6 Challenges and Open Problems

The integration of Agentic AI in healthcare is a promising advancement that aims to improve medical reasoning, diagnosis, and decision-making through multi-agent collaboration. However, despite recent progress, several challenges remain that hinder its full adoption in clinical practice. In this section, we summarize the main challenges facing the field of Agentic AI for healthcare and discuss some open problems.

Failures and Limitations in Healthcare AI Agents: Recent analyses of autonomous capability scaling suggest framing reliability in terms of time-horizon: the length of tasks an agent can complete with $\geq 50\%$ success [56], which is growing but remains short relative to many clinical workflows (multi-hour, multi-step), implying compounding error risk in care pathways²². In realistic clinical simulations, agentic performance drops sharply compared with static QA: AgentClinic [101] reports order-of-magnitude declines in diagnostic accuracy under interactive, multi-modal cases, and AI Hospital [22] finds persistent gaps in multi-turn interviewing/diagnosis despite strong single-step scores—underscoring long-horizon brittleness in healthcare settings. Beyond accuracy, safety and behavior remain limited: ToolEmu shows that 69% of emulator-identified failures manifest with real tools [99], while R-Judge finds leading agents underperform humans on risk awareness, indicating overconfidence and poor self-assessment [137]. Healthcare-specific hallucination and bias are documented in Med-HALT and human-rated clinical studies (e.g., Med-PaLM 2), highlighting plausible-but-wrong outputs and non-uniform safety across populations [81, 108]. Finally, security exposures—from RAG poisoning (PoisonedRAG [147]) to malfunction-amplification attacks on agents (Breaking Agents [139])—demonstrate that even “tool-using” guardrails can fail without provenance checks, allow-lists, and isolation.

Ethical Considerations: Ethical considerations are at the core of AI agent development and deployment in healthcare, where decisions can directly impact patient well-being and medical outcomes. Autonomous AI agents must adhere to principles of fairness, transparency, accountability, and privacy, particularly when handling sensitive medical data, assisting in diagnostics, or supporting clinical decision-making. Bias in AI models is one of the primary ethical concerns, as it can lead to disparities in healthcare recommendations and disproportionately affect underrepresented groups. AI agents with tool-calling capabilities, such as accessing EHRs or generating

²²<https://metr.org/measuring-autonomous-ai-capabilities/>

Table 5. Datasets and Benchmarks for Agentic AI in Healthcare

Dataset	Domain	Purpose	Clinical Utility / Real-World Use Case
MedQA	Medical Question Answering	Multiple-choice medical questions from real-world exams (US, China, Taiwan)	Tests foundational medical knowledge; useful for evaluating diagnostic reasoning but lacks context of real patients or EHR environments.
PubMedQA	Medical Question Answering	Biomedical QA sourced from PubMed abstracts	Suited for literature-grounded reasoning. Not aligned with patient-level tasks or real-time decision support.
JAMA	Medical Question Answering	Clinical questions from JAMA articles	More clinically grounded than exam QA, but lacks patient context or multimodal reasoning.
Medbullets	Medical Question Answering	Multiple-choice medical questions from open-access forums	Designed for education; not based on real patient data or clinical documentation.
DDXPlus	Diagnostic Reasoning	Synthetic patient data for differential diagnosis tasks	Good for structured diagnostic reasoning but lacks fidelity to actual clinical records or unstructured notes.
SymCat	Diagnostic Reasoning	Symptom-condition associations for diagnosis prediction	Based on patient-reported symptoms; useful for prototyping symptom checkers but limited clinical realism.
PMC-VQA	Medical Visual Interpretation	VQA for biomedical images from research articles	Benchmark for vision-language reasoning; does not reflect real radiology data or workflows.
PathVQA	Medical Visual Interpretation	VQA dataset based on pathology images	Educational dataset for pathology QA; not integrated with patient records or diagnostic pipelines.
MedVidQA	Medical Video Interpretation	Video-based medical QA from YouTube tutorials	Useful for patient education and training simulations; lacks standardization and clinical metadata.
MIMIC-CXR	Medical Imaging	Large-scale chest X-ray dataset with QA labels	Derived from real ICU patients; highly applicable to clinical imaging agents and decision support systems.

patient treatment plans, must strictly comply with medical data protection regulations (HIPAA, GDPR, etc.). Unauthorized access, data leaks, or improper decision-making by autonomous agents could expose sensitive patient information, leading to potential legal and ethical violations.

Limited Medical Knowledge in LLMs: LLMs are primarily trained on general-purpose datasets and lack deep, specialized medical expertise which directly influence the agents behavior. Restricted access to medical training data due to privacy concerns and the high cost of curating specialized datasets limits the amount of medical data available for training. LLMs often generate inaccurate or misleading medical information, leading to potentially dangerous conclusions. Medical-specific LLMs, such as Med-PaLM [107, 108], exist but are proprietary and not widely accessible for research and development.

Multi-Agent Collaboration Optimization: Determining the optimal number and types of AI agents for specific medical tasks remains unsolved [54, 115]. Balancing agent diversity is a challenge, as different medical specialties, such as cardiology and radiology, require distinct knowledge bases, making comprehensive collaboration difficult. Increasing the number of agents enhances decision-making but also introduces inefficiencies and computational

Table 6. A structured view of the key evaluation criteria for Agentic AI frameworks, particularly in healthcare applications ensuring robustness, security and efficiency.

Category	Metric	Description
Performance Evaluation	Accuracy, Precision, Recall, F1-score	Standard ML metrics to measure classification or task performance [52]
	Success Rate	Measures how often an agent completes a task correctly [65]
	Output Similarity	Compares generated responses to human responses for quality assessment
	Memory Persistence	Evaluates how well an agent retains relevant information over time [89]
	Efficiency (Response Time, Resource Usage)	Measures computational cost, speed, and efficiency
	Tool Calling Efficiency	Evaluates how accurately and effectively an agent utilizes external tools [99]
Security	Task Scalability	Assesses how well an agent handles increasing workload or complexity [120]
	Confidentiality, Integrity, Availability	Standard security metrics ensuring data protection and system resilience [41, 53]
	Resilience to Adversarial Attacks	Measures robustness against security threats or model manipulation [41, 53]
Privacy	Access Control & Explainability	Ensures proper user authentication and system interpretability [41, 53]
	Differential Privacy	Evaluates how well an agent protects sensitive information from re-identification [12]
	Privacy Leakage Quantification	Measures how much sensitive data can be inferred from agent outputs [12]
Task Adherence	Information Flow Analysis	Assesses how data is shared and controlled across multi-agent systems [99]
	Task Completion Success Rate	Tracks whether the agent achieves its goals without deviation [42, 137]
	Adherence to Task Priorities	Measures how well the agent respects pre-defined priority tasks
Self-Behavior & Adaptation	Recovery Time from Distractions	Evaluates how efficiently an agent refocuses on its goal after interruptions
	Adaptation Success Rate	Evaluates how well an agent learns and improves over time [6]
	Model Accuracy Over Time	Measures how performance changes with ongoing learning
	Resource Efficiency	Tracks how well an agent optimizes computational and memory resources
	Convergence Speed	Measures how quickly an agent adapts and stabilizes during learning [42]

overhead. Multi-agent setups often yield conflicting answers, necessitating robust arbitration and consensus-building mechanisms. Therefore, determine when collaboration is necessary and when a single agent is sufficient is a challenging task for Agentic AI systems in healthcare.

Real-World Clinical Workflow Integration: Current AI agent frameworks do not align well with real-world medical workflows, limiting their practical utility. [22] highlights the challenge of making AI-powered patients realistically uncertain while ensuring that AI doctors ask appropriate follow-up questions. Many medical AI systems are optimized for text-based reasoning but struggle with integrating other data such as medical images, lab tests, and patient histories.

Efficiency vs. Accuracy Trade-off: Multi-agent AI setups introduce significant computational costs that must be balanced against diagnostic accuracy. Running multiple agents in real-time clinical settings is costly and may not be feasible for routine healthcare applications. For example, to run experiments on the entire test sets using GPT-4 (Vision), the authors of the MDAgents framework estimated the total cost to be around \$172,705 [54]. Similarly, EHRAgent [105] incurs notable costs—up to \$0.60 per query on large datasets like MIMIC-III—due to its multi-turn code generation and iterative debugging cycles. TriageAgent [69], despite showing improved ESI-level classification, requires substantial API calls (up to 600+ per test set) and incurs 1.3+ seconds per case, making it difficult to scale in real-time emergency triage. Lastly, hospitals and clinics often have limited computational resources and tight latency requirements, further complicating the deployment of large-scale multi-agent AI systems. These trade-offs highlight the need for cost-effective, low-latency Agentic AI architectures tailored to clinical constraints.

Multimodal Data Fusion in Agentic AI: Integrating multimodal data remains a fundamental yet underexplored challenge in deploying Agentic AI systems for healthcare. Clinical environments often involve a fusion of unstructured text (e.g., progress notes), structured EHRs, medical images (e.g., X-rays, CT scans), and increasingly, video-based content (e.g., ultrasound, surgical footage). While datasets such as PMC-VQA and MedVidQA provide valuable benchmarks, they also expose key limitations. One issue is the heterogeneity of modalities, each with distinct data representations, temporal dynamics, and semantic density—posing difficulties for unified encoding or fusion strategies. Moreover, temporal misalignment between modalities, especially in video-based QA, leads to ambiguous supervision and degraded agent performance. There is also the problem of modality imbalance, where

models tend to overfit to dominant text-based patterns while underutilizing rich but complex visual features. Existing architectures often rely on early fusion (joining embeddings at the input level) or late fusion (aggregating outputs after independent processing), but both approaches introduce trade-offs in cross-modal reasoning and efficiency. In Agentic settings, coordinating across multiple specialized agents further compounds this challenge, requiring synchronization protocols, shared memory, and modality-aware attention. Finally, the scarcity of large, well-annotated multimodal datasets in healthcare, combined with the high cost of curation, limits the ability to train robust and generalizable Agentic models. Addressing these challenges is critical for building AI agents capable of holistic clinical reasoning across complex, multimodal input spaces.

Explainability and Accountability: AI-generated medical decisions require transparency and interpretability to ensure trust and regulatory compliance. Multi-agent AI systems aggregate knowledge from various agents, making it difficult to pinpoint responsibility for errors. Without clear explainability, AI-driven medical errors may lead to liability issues and resistance from healthcare providers. Clinicians are unlikely to rely on AI agents unless they can fully understand and justify the AI’s recommendations.

Reasoning Failures in Agentic AI: Recent evaluations of LLM-based agents have revealed persistent failure modes that are especially critical to address in high-stakes domains like healthcare. Sun et al. [112] introduced the OMEGA benchmark, which probes the generalization capabilities of leading LLMs across increasingly complex reasoning tasks. Their analysis identifies key vulnerabilities, including: (1) *overthinking and recursive error spirals*, where agents generate excessive reasoning steps that introduce mistakes even after reaching the correct answer; (2) *heuristic guessing*, where agents substitute computation with speculative answers under increased complexity; and (3) *limited compositional and transformative reasoning*, where agents fail to combine known skills or adapt their strategies to novel task structures. These behaviors raise serious concerns for clinical applications, where hallucinated reasoning, skipped safety checks, or brittle workflows could lead to harmful diagnostic or treatment decisions. For example, an agent may initially identify a correct triage path but “reason itself out of it” by introducing irrelevant justifications, or it may apply memorized treatment patterns without verifying current vitals or lab results. Addressing these failure modes requires not only better model alignment and training data, but also architectural safeguards, such as memory-grounded verification, structured oversight, and human-in-the-loop escalation—especially before deploying Agentic AI systems in real-world healthcare environments.

Compounding Error in Multi-Step Clinical Workflows. Unlike single-turn QA benchmarks where most healthcare agents are evaluated, real clinical workflows involve sequential, interdependent decisions, history-taking, differential diagnosis, test ordering, interpretation, and treatment planning. Errors at any stage propagate and amplify downstream. AgentClinic [101] demonstrates order-of-magnitude accuracy drops in interactive settings compared to static QA, yet no existing framework formally models or mitigates this compounding effect. A critical open problem is developing error-aware planning mechanisms where agents can quantify cumulative uncertainty across a reasoning chain and trigger escalation before downstream decisions are contaminated. This connects to recent findings on autonomous task time-horizons [56], which show that agent reliability degrades sharply as task length increases; a particularly concerning pattern for multi-hour clinical pathways.

Calibration–Autonomy Mismatch. Confidence-based escalation (as proposed in Section 3.4) assumes that agent confidence scores are well-calibrated. In practice, LLM-based agents exhibit systematic overconfidence, as documented by R-Judge [137] where leading agents underperform humans on risk-awareness tasks. In healthcare, this creates a dangerous failure mode: an agent that is confidently wrong will bypass escalation thresholds and execute harmful actions autonomously. The open problem is twofold: (1) developing calibration methods specifically for agentic reasoning chains (not just single-turn outputs), and (2) designing escalation mechanisms that are robust to miscalibration, such as uncertainty ensembles or independent verification agents.

Conclusion

Agentic AI is poised to revolutionize healthcare by enabling autonomous, adaptive, and collaborative AI systems capable of performing complex medical tasks with minimal human intervention. While existing single-agent and multi-agent frameworks have demonstrated significant advancements in areas such as clinical decision support, rare disease diagnosis, and EHR automation, several challenges remain. Trust, explainability, security, and regulatory compliance are critical factors that must be addressed for successful integration into clinical workflows. Moving forward, the field must focus on scalable and interpretable AI architectures, ethical deployment frameworks, and robust validation methodologies to ensure that Agentic AI delivers safe, reliable, and equitable improvements to global healthcare systems.

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Received 21 March 2025; revised 18 March 2026; accepted 6 April 2026